Makhura Kamogelo

Week 6 practical

Tasks:

Step 1: Configure eBGP

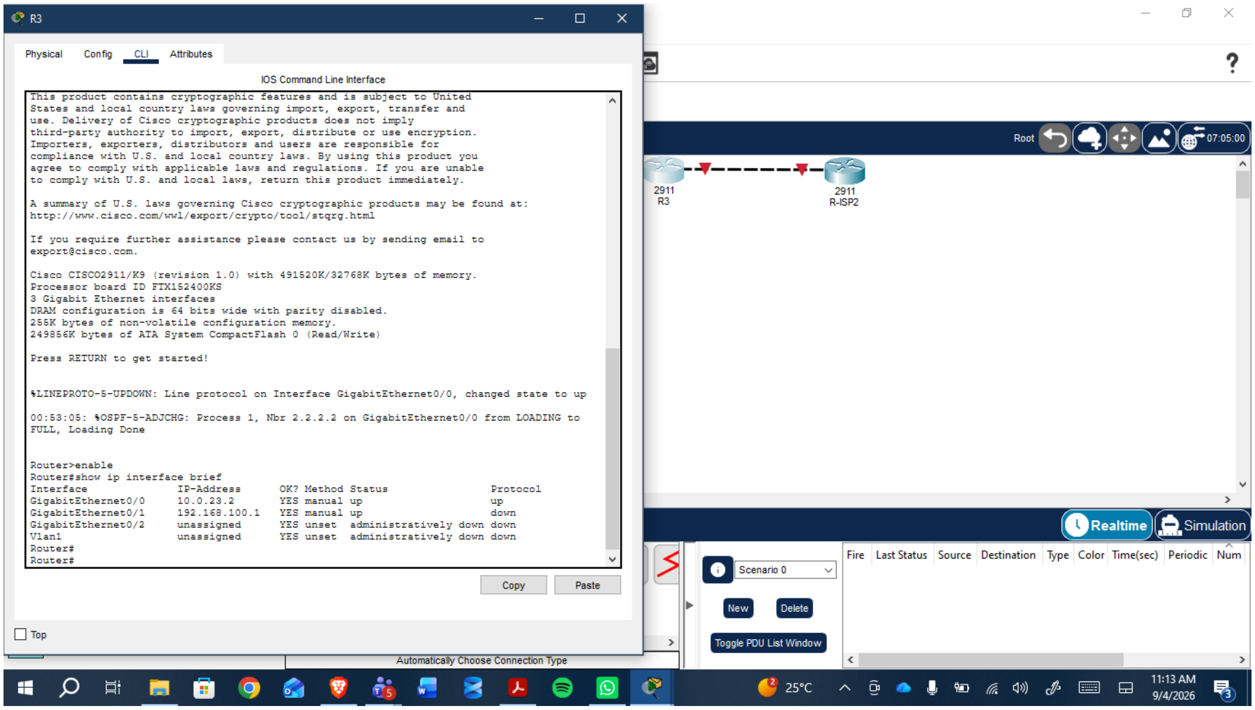
1. On R3 (AS 65001): router bgp 65001 neighbor [R-ISP2's IP] remote-as 65002
2. On R-ISP2 (AS 65002): router bgp 65002 neighbor [R3's IP] remote-as 65001

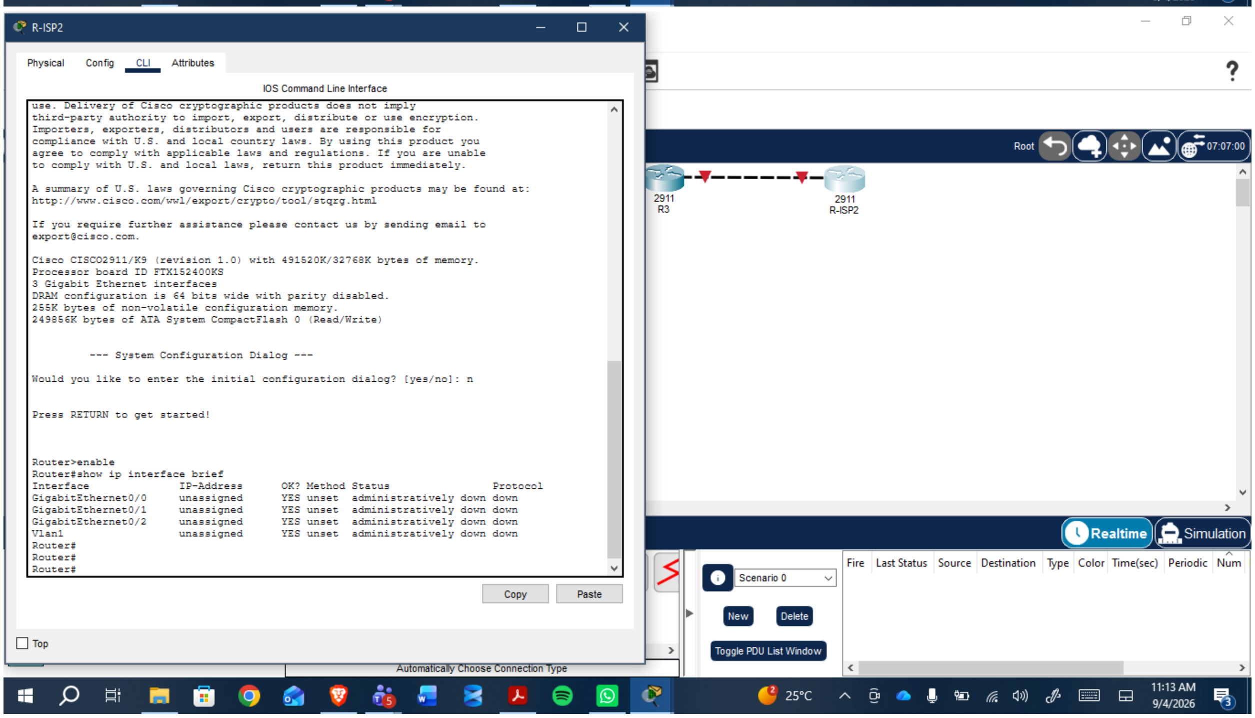
Step 2: Advertise Networks

1. On R3: network [summarized internal network] mask [mask]
2. On R-ISP2: network [its own network] mask [mask]

Step 3: Verify

1. show ip bgp summary
2. show ip route bgp
3. Configure loopback interfaces on each side and ping between them to confirm reachability.


R3
Physical
Config
CLI
Attributes
IOS Command Line Interface
This product contains cryptographic features and is subject to United
States and local country laws governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.
A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
If you require further assistance please contact us by sending email to
export@cisco.com.
Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)
Press RETURN to get started!
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
00:53:05: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to
FULL, Loading Done
Router>enable
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.0.23.2 YES manual up up
GigabitEthernet0/1 192.168.100.1 YES manual up down
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
Router#
Copy
Paste
Top
Root
07:05:00
2911
R3
2911
R-ISP2
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Automatically Choose Connection Type
25°C
11:13 AM
9/4/2026


R-ISP2
Physical
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CLI
Attributes
IOS Command Line Interface
use. Delivery of Cisco cryptographic products does not imply
third-party authority to import, export, distribute or use encryption.
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249856K bytes of ATA System CompactFlash 0 (Read/Write)
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: n
Press RETURN to get started!
Router>enable
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES unset administratively down down
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
Router#
Router#
Copy
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Top
Root
07:07:00
2911
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R-ISP2
Realtime
Simulation
Scenario 0
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Toggle PDU List Window
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Last Status
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Destination
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Color
Time(sec)
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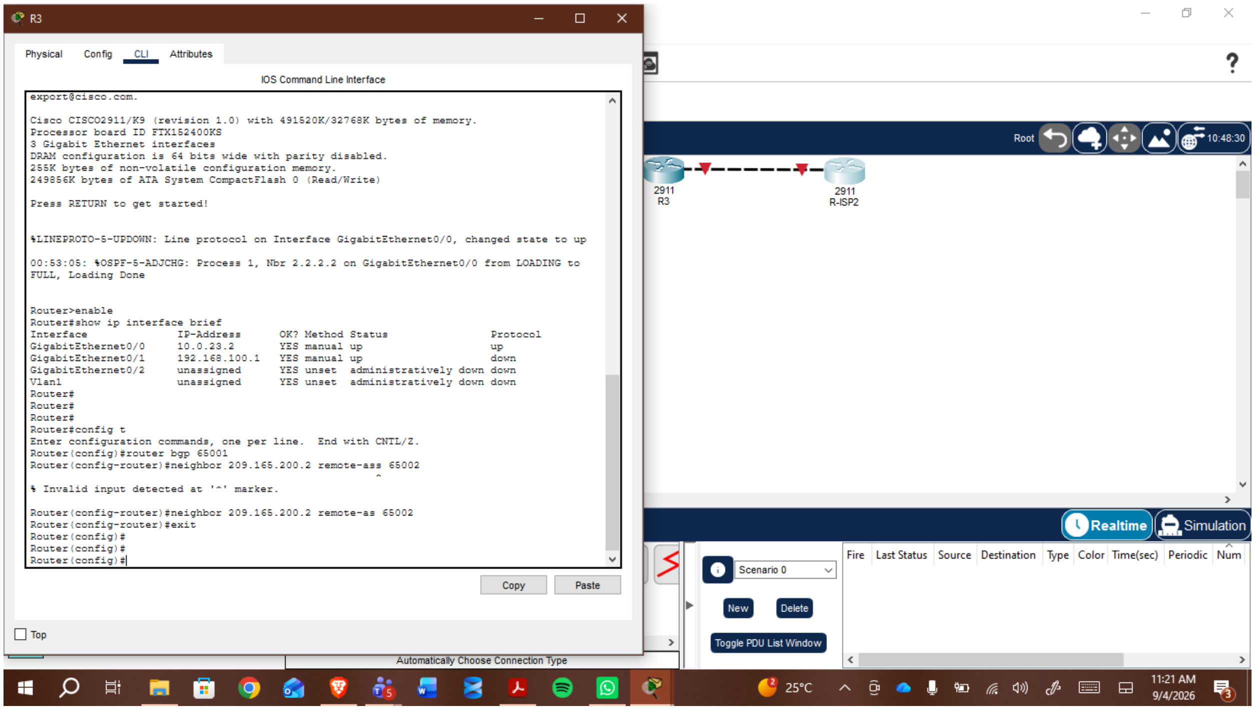
Num
Automatically Choose Connection Type
25°C
11:13 AM
9/4/2026


R3
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Attributes
IOS Command Line Interface
export@cisco.com.
Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)
Press RETURN to get started!
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
00:53:05: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
Router>enable
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.0.23.2 YES manual up up
GigabitEthernet0/1 192.168.100.1 YES manual up down
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
Router#
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 65001
Router(config-router)#neighbor 209.165.200.2 remote-ass 65002
% Invalid input detected at '^' marker.
Router(config-router)#neighbor 209.165.200.2 remote-as 65002
Router(config-router)#exit
Router(config)#
Router(config)#
Router(config)#
Copy
Paste
Top
2911
R3
2911
R-ISP2
Root
10:48:30
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
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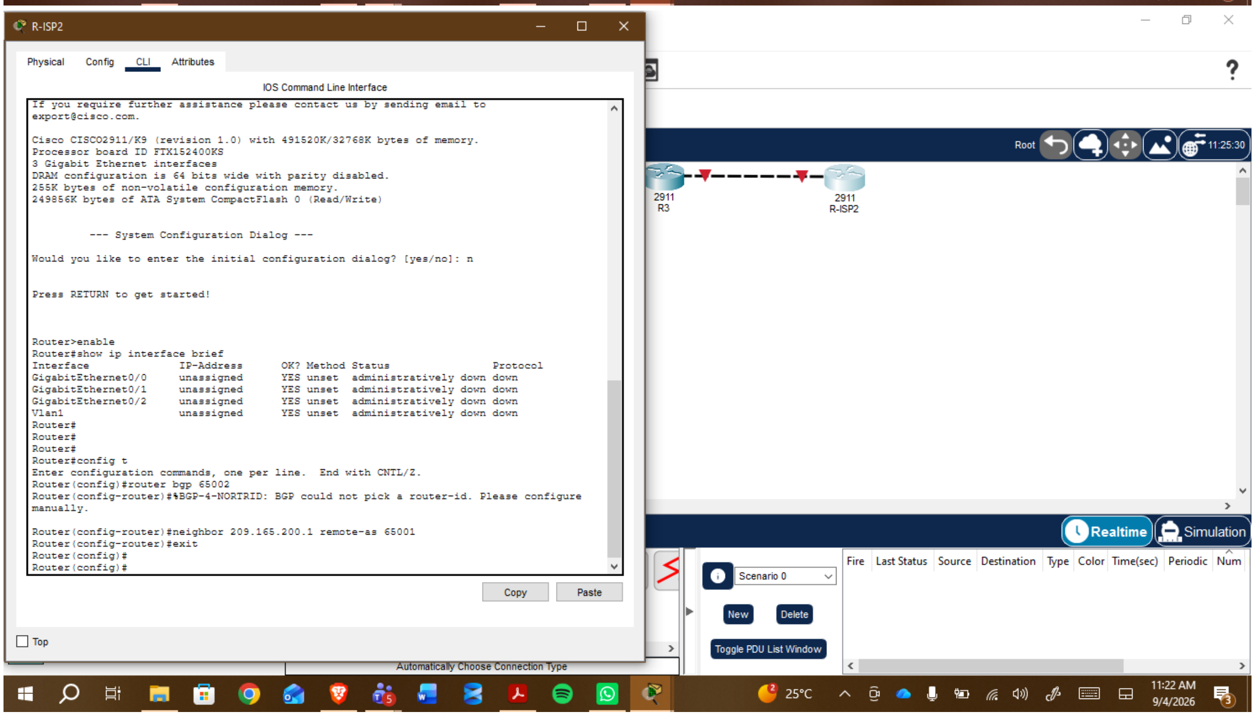
R-ISP2
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export@cisco.com.
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Press RETURN to get started!
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Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES unset administratively down down
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
Router#
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 65002
Router(config-router)#%BGP-4-NORTRID: BGP could not pick a router-id. Please configure manually.
Router(config-router)#neighbor 209.165.200.1 remote-as 65001
Router(config-router)#exit
Router(config)#
Router(config)#
Copy
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Top
2911
R3
2911
R-ISP2
Root
11:25:30
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
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Time(sec)
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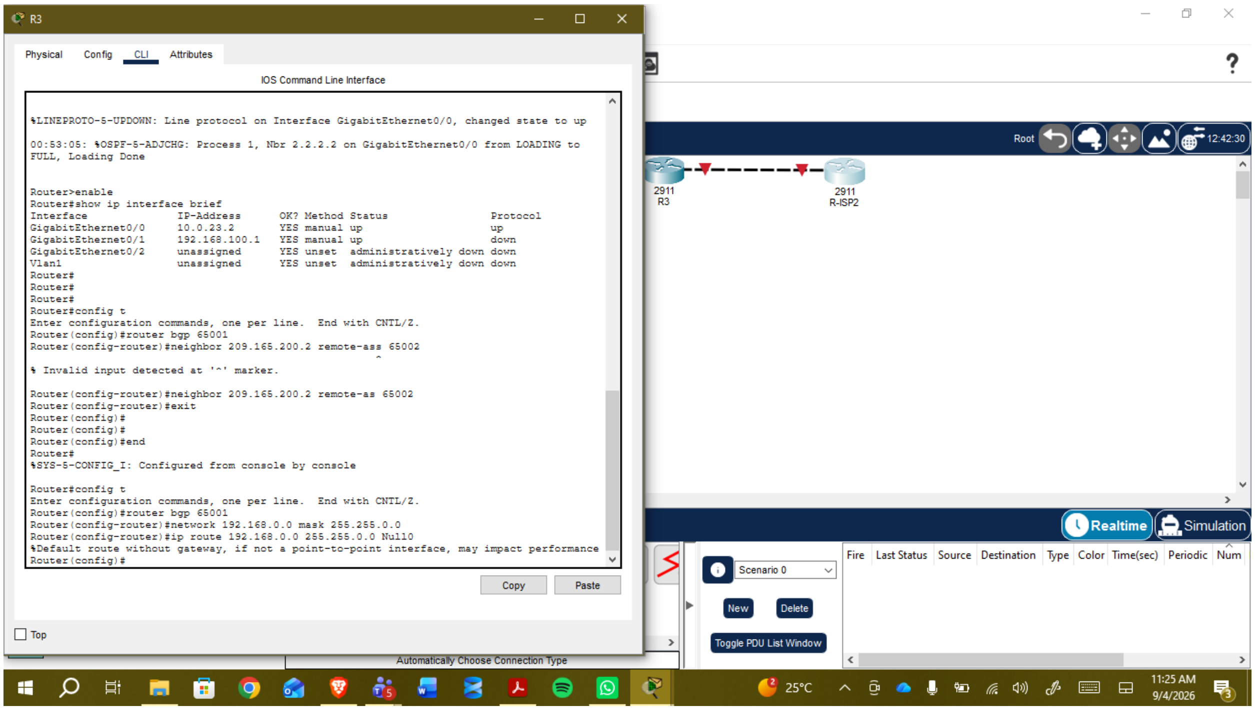
Automatically Choose Connection Type
25°C
11:22 AM
9/4/2026


R3
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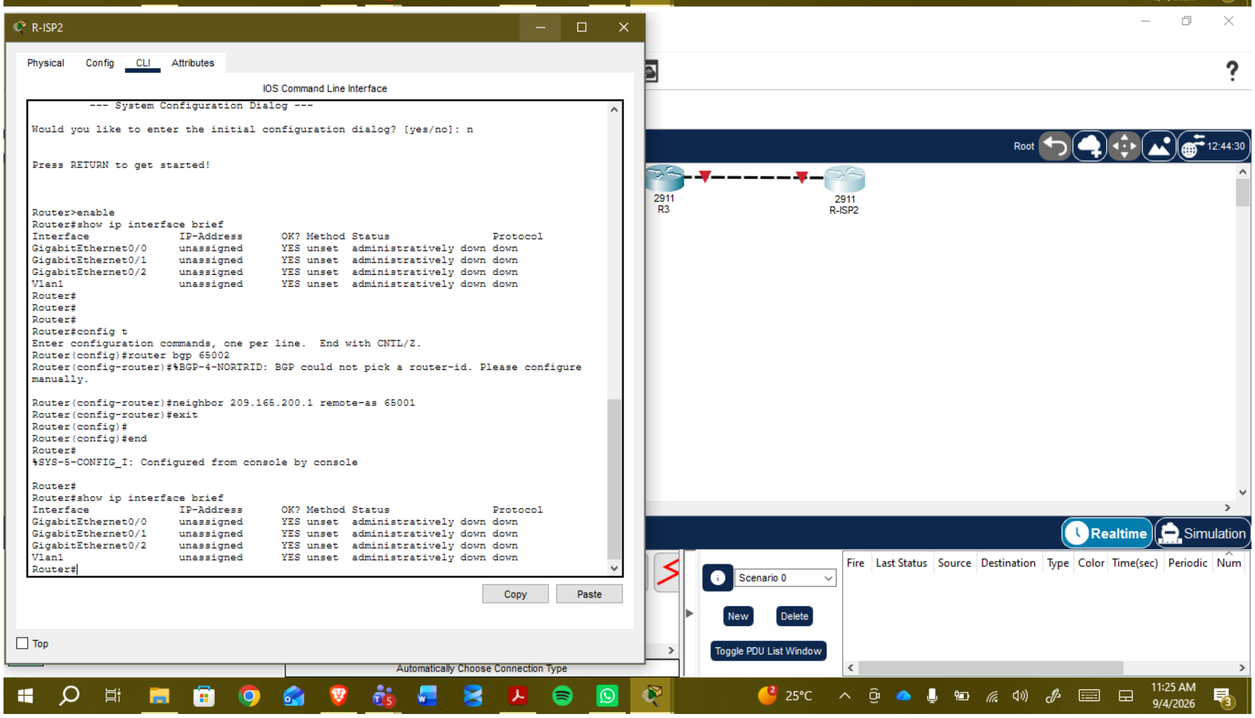
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
00:53:05: %OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on GigabitEthernet0/0 from LOADING to FULL, Loading Done
Router>enable
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 10.0.23.2 YES manual up up
GigabitEthernet0/1 192.168.100.1 YES manual up down
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
Router#
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 65001
Router(config-router)#neighbor 209.165.200.2 remote-ass 65002
^
% Invalid input detected at '^' marker.
Router(config-router)#neighbor 209.165.200.2 remote-as 65002
Router(config-router)#exit
Router(config)#
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 65001
Router(config-router)#network 192.168.0.0 mask 255.255.0.0
Router(config-router)#ip route 192.168.0.0 255.255.0.0 Null0
%Default route without gateway, if not a point-to-point interface, may impact performance
Router(config)#
Copy
Paste
Top
Automatically Choose Connection Type
Root
12:42:30
2911
R3
2911
R-ISP2
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
25°C
11:25 AM
9/4/2026


R-ISP2
Physical
Config
CLI
Attributes
IOS Command Line Interface
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: n
Press RETURN to get started!
Router>enable
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES unset administratively down down
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
Router#
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
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Router(config-router)#exit
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 unassigned YES unset administratively down down
GigabitEthernet0/1 unassigned YES unset administratively down down
GigabitEthernet0/2 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
Router#
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Paste
Top
Automatically Choose Connection Type
Root
12:44:30
2911
R3
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R-ISP2
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
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Time(sec)
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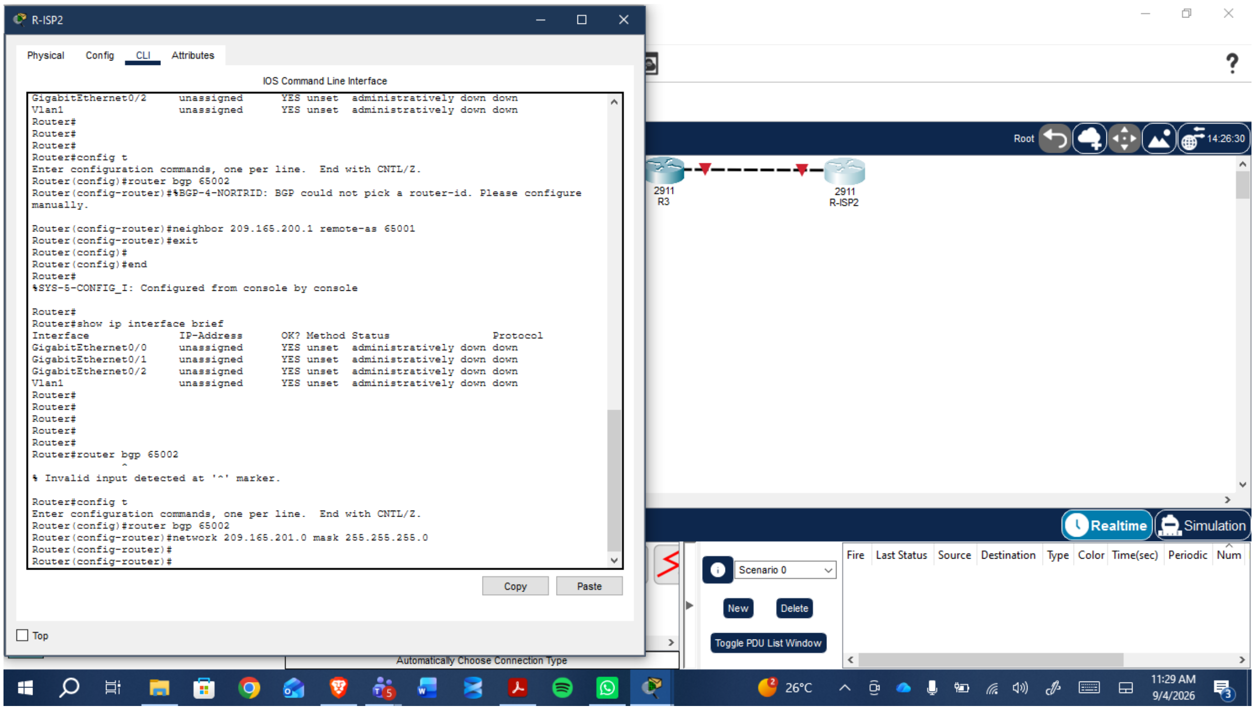
25°C
11:25 AM
9/4/2026


R-ISP2
Physical
Config
CLI
Attributes
IOS Command Line Interface
GigabitEthernet0/2   unassigned   YES unset administratively down down
Vlan1   unassigned   YES unset administratively down down
Router#
Router#
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 65002
Router(config-router)#%BGP-4-NORTRID: BGP could not pick a router-id. Please configure manually.
Router(config-router)#neighbor 209.165.200.1 remote-as 65001
Router(config-router)#exit
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#show ip interface brief
Interface   IP-Address   OK? Method Status   Protocol
GigabitEthernet0/0   unassigned   YES unset administratively down down
GigabitEthernet0/1   unassigned   YES unset administratively down down
GigabitEthernet0/2   unassigned   YES unset administratively down down
Vlan1   unassigned   YES unset administratively down down
Router#
Router#
Router#
Router#
Router#
Router#router bgp 65002
% Invalid input detected at '^' marker.
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 65002
Router(config-router)#network 209.165.201.0 mask 255.255.255.0
Router(config-router)#
Router(config-router)#
Copy
Paste
Top
Automatically Choose Connection Type
Root
14:26:30
2911
R3
2911
R-ISP2
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
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Last Status
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Time(sec)
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26°C
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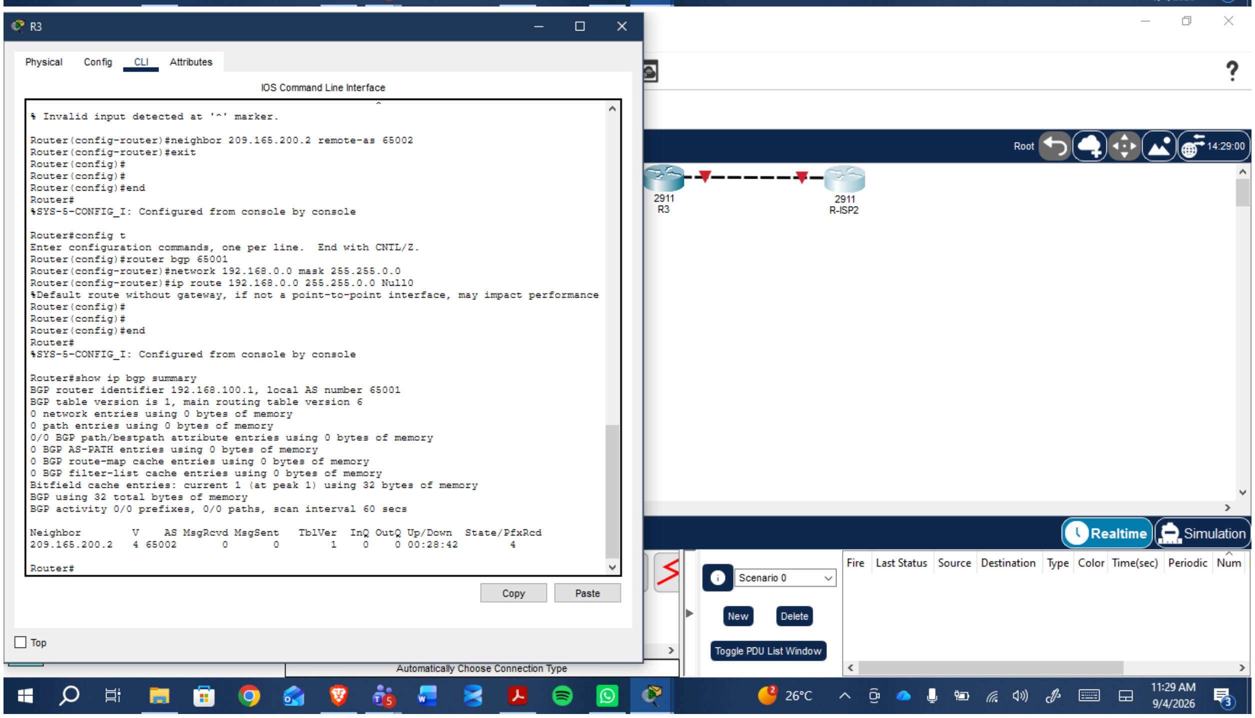
9/4/2026


R3
Physical
Config
CLI
Attributes
IOS Command Line Interface
% Invalid input detected at '^' marker.
Router(config-router)#neighbor 209.165.200.2 remote-as 65002
Router(config-router)#exit
Router(config)#
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 65001
Router(config-router)#network 192.168.0.0 mask 255.255.0.0
Router(config-router)#ip route 192.168.0.0 255.255.0.0 Null0
%Default route without gateway, if not a point-to-point interface, may impact performance
Router(config)#
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#show ip bgp summary
BGP router identifier 192.168.100.1, local AS number 65001
BGP table version is 1, main routing table version 6
0 network entries using 0 bytes of memory
0 path entries using 0 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
0 BGP AS-PATH entries using 0 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 32 total bytes of memory
BGP activity 0/0 prefixes, 0/0 paths, scan interval 60 secs
Neighbor   V   AS MsgRcvd MsgSent   TblVer InQ OutQ Up/Down State/PfxRcd
209.165.200.2   4 65002   0   0   1   0   0 00:28:42   4
Router#
Copy
Paste
Top
Automatically Choose Connection Type
Root
14:29:00
2911
R3
2911
R-ISP2
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
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Time(sec)
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26°C
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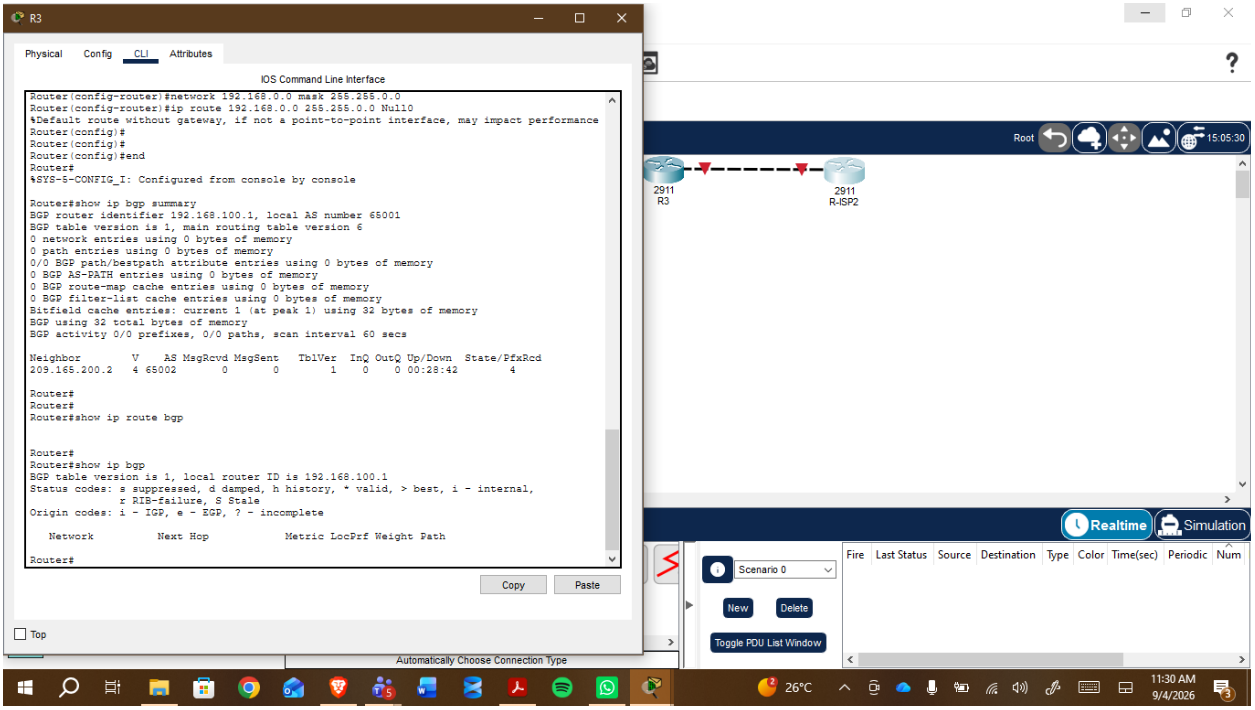
9/4/2026


R3
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router(config-router)#network 192.168.0.0 mask 255.255.0.0
Router(config-router)#ip route 192.168.0.0 255.255.0.0 Null0
%Default route without gateway, if not a point-to-point interface, may impact performance
Router(config)#
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#show ip bgp summary
BGP router identifier 192.168.100.1, local AS number 65001
BGP table version is 1, main routing table version 6
0 network entries using 0 bytes of memory
0 path entries using 0 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
0 BGP AS-PATH entries using 0 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 32 total bytes of memory
BGP activity 0/0 prefixes, 0/0 paths, scan interval 60 secs
Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State/PfxRcd
209.165.200.2 4 65002 0 0 1 0 0 00:28:42 4
Router#
Router#
Router#show ip route bgp
Router#
Router#show ip bgp
BGP table version is 1, local router ID is 192.168.100.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete
Network Next Hop Metric LocPrf Weight Path
Router#
Copy
Paste
Top
Automatically Choose Connection Type
Root
15:05:30
2911
R3
2911
R-ISP2
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
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Time(sec)
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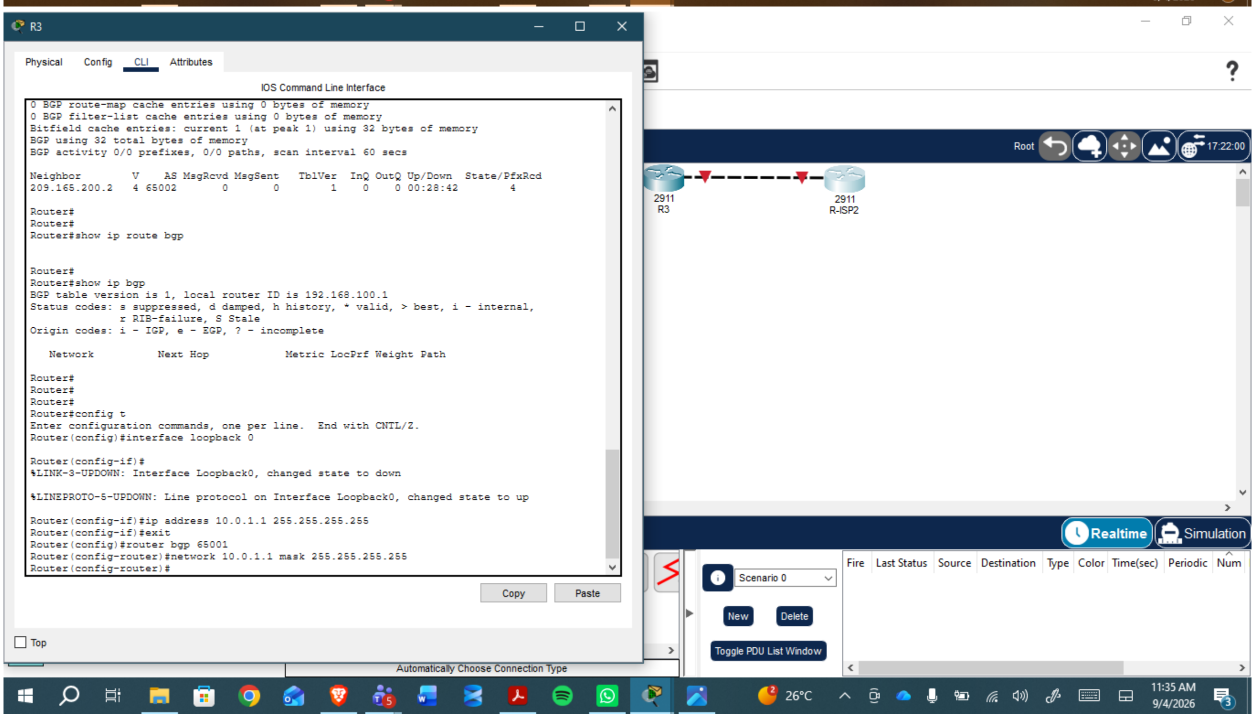
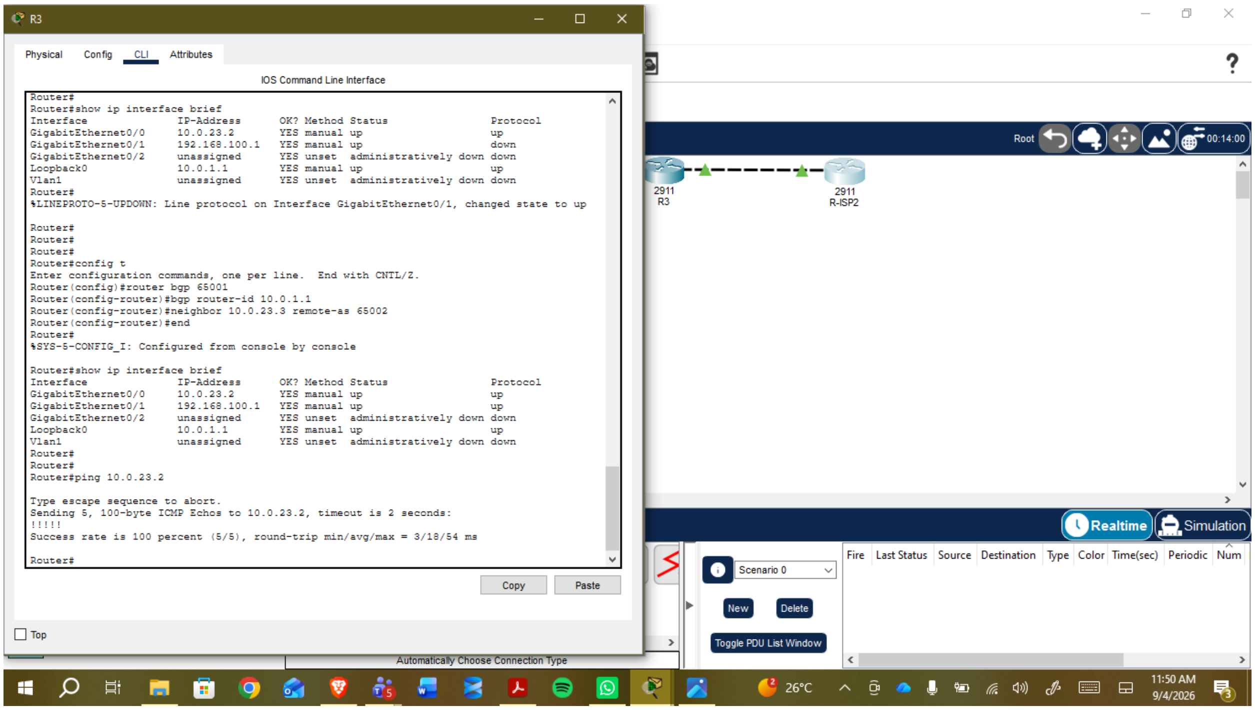
Num
26°C
11:30 AM
9/4/2026


R3
Physical
Config
CLI
Attributes
IOS Command Line Interface
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 32 total bytes of memory
BGP activity 0/0 prefixes, 0/0 paths, scan interval 60 secs
Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State/PfxRcd
209.165.200.2 4 65002 0 0 1 0 0 00:28:42 4
Router#
Router#
Router#show ip route bgp
Router#
Router#show ip bgp
BGP table version is 1, local router ID is 192.168.100.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete
Network Next Hop Metric LocPrf Weight Path
Router#
Router#
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface loopback 0
Router(config-if)#
%LINK-3-UPDOWN: Interface Loopback0, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
Router(config-if)#ip address 10.0.1.1 255.255.255.255
Router(config-if)#exit
Router(config)#router bgp 65001
Router(config-router)#network 10.0.1.1 mask 255.255.255.255
Router(config-router)#
Copy
Paste
Top
Automatically Choose Connection Type
Root
17:22:00
2911
R3
2911
R-ISP2
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
26°C
11:35 AM
9/4/2026


R3
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router#
Router#show ip interface brief
Interface   IP-Address   OK? Method Status   Protocol
GigabitEthernet0/0   10.0.23.2   YES manual up   up
GigabitEthernet0/1   192.168.100.1   YES manual up   down
GigabitEthernet0/2   unassigned   YES unset administratively down down
Loopback0   10.0.1.1   YES manual up   up
Vlan1   unassigned   YES unset administratively down down
Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router#
Router#
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 65001
Router(config-router)#bgp router-id 10.0.1.1
Router(config-router)#neighbor 10.0.23.3 remote-as 65002
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#show ip interface brief
Interface   IP-Address   OK? Method Status   Protocol
GigabitEthernet0/0   10.0.23.2   YES manual up   up
GigabitEthernet0/1   192.168.100.1   YES manual up   up
GigabitEthernet0/2   unassigned   YES unset administratively down down
Loopback0   10.0.1.1   YES manual up   up
Vlan1   unassigned   YES unset administratively down down
Router#
Router#
Router#ping 10.0.23.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.23.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 3/18/54 ms
Router#
Copy
Paste
Top
Automatically Choose Connection Type
Root
00:14:00
2911
R3
2911
R-ISP2
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
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Time(sec)
Periodic
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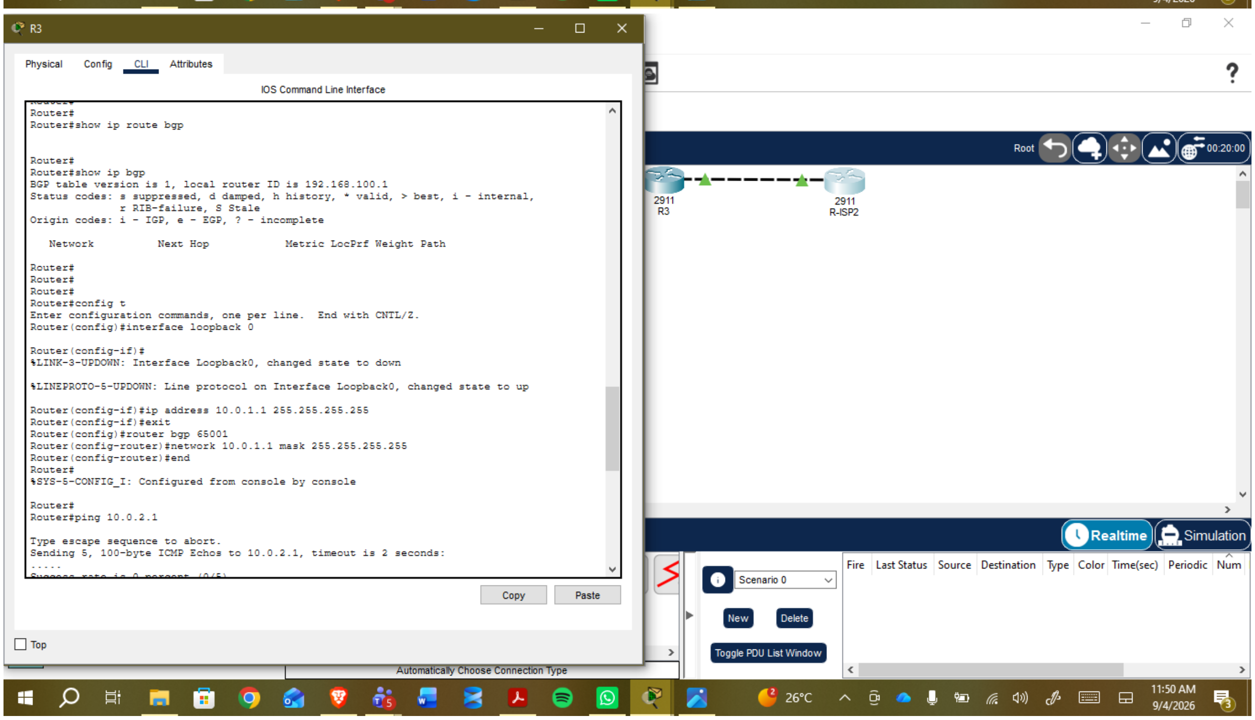
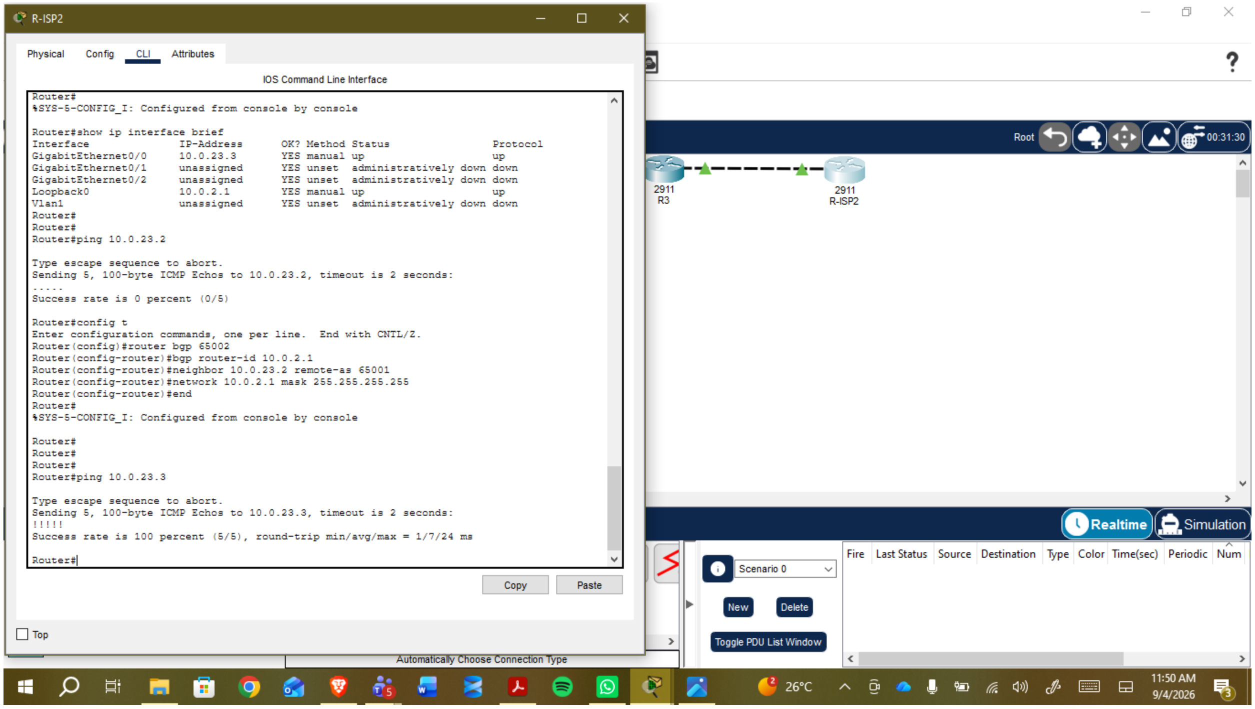
26°C
11:50 AM
9/4/2026


R3
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router#
Router#show ip route bgp
Router#
Router#show ip bgp
BGP table version is 1, local router ID is 192.168.100.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete
   Network   Next Hop   Metric LocPrf Weight Path
Router#
Router#
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface loopback 0
Router(config-if)#
%LINK-3-UPDOWN: Interface Loopback0, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
Router(config-if)#ip address 10.0.1.1 255.255.255.255
Router(config-if)#exit
Router(config)#router bgp 65001
Router(config-router)#network 10.0.1.1 mask 255.255.255.255
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#ping 10.0.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.2.1, timeout is 2 seconds:
.....
Copy
Paste
Top
Automatically Choose Connection Type
Root
00:20:00
2911
R3
2911
R-ISP2
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
26°C
11:50 AM
9/4/2026



R-ISP2
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#show ip interface brief
Interface   IP-Address   OK? Method Status   Protocol
GigabitEthernet0/0   10.0.23.3   YES manual up   up
GigabitEthernet0/1   unassigned   YES unset administratively down down
GigabitEthernet0/2   unassigned   YES unset administratively down down
Loopback0   10.0.2.1   YES manual up   up
Vlan1   unassigned   YES unset administratively down down
Router#
Router#
Router#ping 10.0.23.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.23.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router bgp 65002
Router(config-router)#bgp router-id 10.0.2.1
Router(config-router)#neighbor 10.0.23.2 remote-as 65001
Router(config-router)#network 10.0.2.1 mask 255.255.255.255
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#
Router#
Router#ping 10.0.23.3
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.23.3, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/7/24 ms
Router#
Copy
Paste
Top
Root
00:31:30
2911
R3
2911
R-ISP2
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Automatically Choose Connection Type
26°C
11:50 AM
9/4/2026

Reflection Questions:

Why would two separate organisations use BGP between themselves instead of just running OSPF across both networks?

- **Trust & Control:** OSPF shares your entire internal topology - every router, every link, every VLAN. BGP only shares what you WANT to share - just the summarized network. You keep internal details hidden.
- **Scalability:** OSPF keeps a full link-state database of the whole internet if you tried. Impossible. BGP is designed for 100,000+ networks.
- **Policy:** BGP lets you control WHAT to advertise and WHAT path to prefer using attributes like AS-PATH, Local Preference, MED. OSPF just picks shortest path, no policy. ISP needs to filter customers.
- **Different Administrations:** OSPF requires one administrative domain. Two companies can't agree on Areas, Router-IDs, authentication. BGP is the standard for inter-AS - eBGP is made for this.